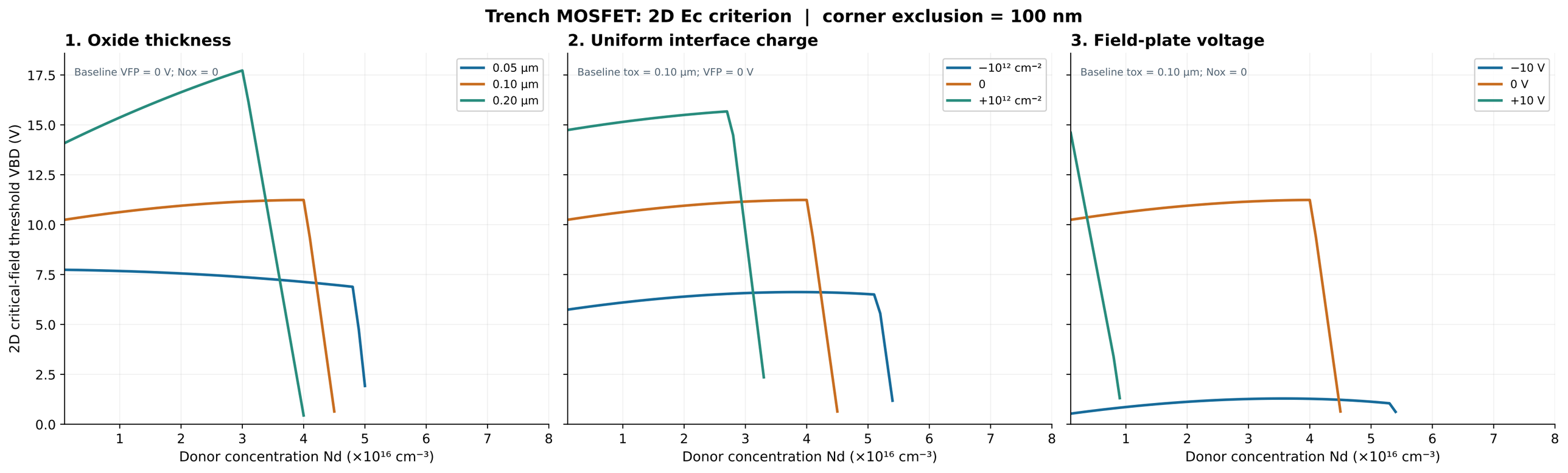

Trench MOSFET: 2D Ec criterion | corner exclusion = 100 nm
1. Oxide thickness
2. Uniform interface charge
3. Field-plate voltage
2D critical-field threshold VBD (V)
Donor concentration Nd (×10¹⁶ cm⁻³)
Baseline VFP = 0 V; Nox = 0
0.05 µm
0.10 µm
0.20 µm
Baseline tox = 0.10 µm; VFP = 0 V
−10¹² cm⁻²
0
+10¹² cm⁻²
Baseline tox = 0.10 µm; Nox = 0
−10 V
0 V
+10 V